

COMPUTER NETWORKS

Ex.No:16	NMAP TO DISCOVER LIVE HOSTS USING ARP, ICMP, TCP AND UDP PING SCANS IN TRYHACKME
DATE:	

Aim: To discover live hosts in a network using Nmap by performing ARP scan, ICMP scan, TCP ping scan, and UDP ping scan through a free TryHackMe learning environment.

Software Requirements:

- Computer or laptop with an internet connection
- Web browser
- Free TryHackMe account
- A free-access lab room with an authorized target
- Nmap terminal in the AttackBox or an authorized lab machine

Procedure:

Step 1: Create a free TryHackMe account

Open <https://tryhackme.com> in a web browser. Select Sign Up, register using an available sign-in method, and log in to the account.

Step 2: Select a free room

Visit <https://tryhackme.com/module/nmap> and look for a room or activity related to Nmap and host discovery. Confirm that it is available under the free plan. If the relevant activity requires a subscription, choose another free room that provides a suitable authorized lab target. The learning-path catalogue is available at <https://tryhackme.com/paths>.

Step 3: Start the lab environment

Open the selected room, read its instructions, and start the target machine. Start the browser-based AttackBox if it is included in the available plan. Note the target IP address shown by the room and open the terminal. If using a local terminal, ensure it can reach the authorized lab target.

Step 4: Verify Nmap installation

Enter the following command in the terminal:

nmap --version

Step 5: Perform ARP scan

Command:

sudo nmap -PR -sn TARGET_IP

Explanation:

- -PR: Uses ARP requests for host discovery.
- -sn: Discovers hosts without performing a port scan.
- TARGET_IP: Replace this placeholder with the IP address of the authorized lab target.

Observation: Record whether Nmap identifies the target as up. ARP discovery generally applies to the same local network segment.

Step 6: Perform ICMP scan

Command:

sudo nmap -PE -sn TARGET_IP

Explanation: -PE sends ICMP Echo Request probes. The -sn option performs host discovery without port scanning.

Observation: Record whether Nmap reports “Host is up”. A firewall may block ICMP requests even when a host is active.

Step 7: Perform TCP ping scan

Command:

sudo nmap -PS80,443 -sn TARGET_IP

Explanation: -PS sends TCP SYN probes to ports 80 and 443. The -sn option disables port scanning.

Observation: Record whether Nmap detects the target as reachable. The result depends on target configuration and firewall rules.

Step 8: Perform UDP ping scan

Command:

sudo nmap -PU53,123 -sn TARGET_IP

Explanation: -PU sends UDP host-discovery probes to ports 53 and 123. The -sn option performs host discovery without port scanning.

Observation: Record the actual output. UDP probes may receive no response even when the target is active.

Observation Table:

S. No.	Scan Technique	Nmap Option	Observed Result
1	ARP scan	-PR	
2	ICMP scan	-PE	
3	TCP SYN ping	-PS	
4	UDP ping	-PU	

Expected observation table

S. No.	Scan technique	Command option	Expected observation
1	ARP scan	-PR	Host detected on the same local network
2	ICMP scan	-PE	Host detected if ICMP response is received
3	TCP SYN ping	-PS	Host detected if a suitable TCP response is received
4	UDP ping	-PU	Host detected if a suitable response is received

Result: Live host discovery was studied using Nmap in the TryHackMe learning environment. ARP, ICMP, TCP, and UDP probes were used to examine host reachability, and the observations were recorded.

Conclusion: Nmap provides multiple techniques for discovering active hosts. The effectiveness of each technique depends on network topology, target configuration, and firewall rules.

Room progress: 50%

- From **computer1**
- To **computer1** (to indicate it is broadcast)
- Packet Type: "ARP Request"
- Data: computer6 (because we are asking for **computer6's** MAC address using ARP Request)

How many devices can see the ARP Request?

4

✓ Correct Answer

Did computer6 receive the ARP Request? (yea/nay)

nay

✓ Correct Answer

Send a packet with the following:

Send Packet

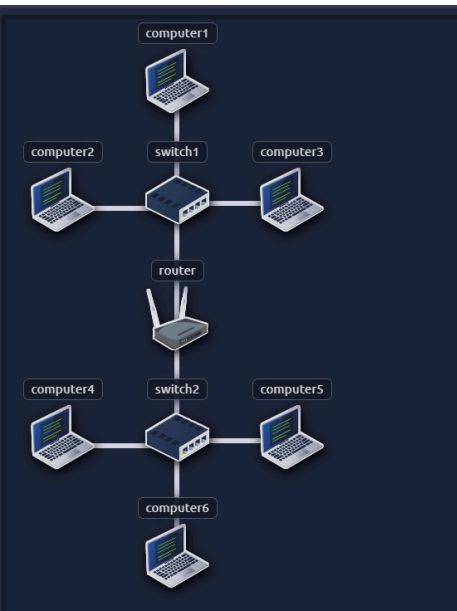
From: computer4

To: computer4

Packet Type: arp_request

Data: computer6

Send Packet



Legend

- TCP Packet
- TCP Handshak

Send Packet

From: computer2

Network Log

ARP REQUEST: Who has computer5 tell router

ARP RESPONSE: Hev

Nmap: Network Simulator

Connecting to the Network

In this room, you will work with a small network consisting of one Windows host and one Ubuntu host to practice identifying live systems on a network.

When you open the room, you will see a network diagram showing the available hosts. To begin, click the **Start** button to deploy the network. Allow up to 2 minutes for all machines to initialise properly. Once the network has finished loading, launch the AttackBox by clicking the **Start AttackBox** button. When the AttackBox opens in split view, open a Terminal and run **Nmap** to perform host discovery, then answer the questions in this room.

Answer the questions below

What is the first IP address Nmap would scan if you provided 10.10.12.13/29 as your target?

10.10.12.8

✓ Correct Answer

How many IP addresses will Nmap scan if you provide the following range: 10.10.0-255.101-125?

6400

✓ Correct Answer

Room progress: 50%

Answer the questions below

Send a packet with the following:
From **computer1**
To **computer3**
Packet Type: "Ping Request"
What type of packet did **computer1** send before the ping?

ARP Request

✓ Correct Answer

What type of packet did **computer1** receive before it was able to send the ping?

ARP Response

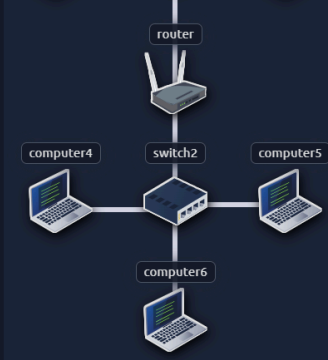
✓ Correct Answer

How many computers responded to the ping request?

1

✓ Correct Answer

Send a packet with the following:
From **computer2**
To **computer5**
Packet Type: "Ping Request"
What is the name of the first device that responded to the first ARP Request?



Legend

- TCP Packet
- TCP Handshak
- UDP Packet
- ARP Packet
- Ping Packet

Send Packet

From: computer2

To: computer5

Packet Type: Ping Request

Data:

Send Packet

Network Log

ARP REQUEST: Who has computer5 tell router

ARP RESPONSE: Hey router, I am computer5

PING: computer5 received ping request from computer2, sending ping response to computer2

PING: Sending Ping Response packet from

Room progress: 77%

Time	Source	Destination	Protocol	Length	Info
7.000000	10.200.5.2	10.200.6.7	ICMP	32	Address mask request
8.000000	10.200.5.2	10.200.6.8	ICMP	32	Address mask request
9.000000	10.200.5.2	10.200.6.9	ICMP	32	Address mask request
10.000000	10.200.5.2	10.200.6.10	ICMP	32	Address mask request
11.000000	10.200.5.2	10.200.6.1	ICMP	32	Address mask request
12.000000	10.200.5.2	10.200.6.2	ICMP	32	Address mask request
13.000000	10.200.5.2	10.200.6.3	ICMP	32	Address mask request
14.000000	10.200.5.2	10.200.6.4	ICMP	32	Address mask request
15.000000	10.200.5.2	10.200.6.5	ICMP	32	Address mask request
16.000000	10.200.5.2	10.200.6.6	ICMP	32	Address mask request
17.000000	10.200.5.2	10.200.6.7	ICMP	32	Address mask request
18.000000	10.200.5.2	10.200.6.8	ICMP	32	Address mask request
19.000000	10.200.5.2	10.200.6.9	ICMP	32	Address mask request
20.000000	10.200.5.2	10.200.6.10	ICMP	32	Address mask request

Frame 1: 32 bytes on wire (256 bits) captured (0.000000 seconds) on interface 0

Raw packet: 0010 0a c8 06 01 11 00 ea ee 04 11 00 00 00 00 00 00

wireshark_tun0INNAK3.pcapng

Packets: 67 · Dropped: 0 (0.0%)

Profile: Default

Answer the questions below

What is the option required to tell Nmap to use ICMP Timestamp to discover live hosts?

-PP

✓ Correct Answer

What is the option required to tell Nmap to use ICMP Address Mask to discover live hosts?

-PM

✓ Correct Answer

What is the option required to tell Nmap to use ICMP Echo to discover live hosts?

-PE

✓ Correct Answer

Room progress: 90%

146	4.869134478	10.200.5.2	10.200.6.250	UDP	28	53835	→	40125	Len=0
147	4.869157659	10.200.5.2	10.200.6.251	UDP	28	53835	→	40125	Len=0

Frame 130:	0000	45 00 00 1c ba aa 00 00	2b 11 b4 32 0a c8 05 02	E.....+..2....
Raw packet:	0010	0a c8 06 63 d2 4b 9c bd	00 08 0f e0	...c.K...o

wireshark tun0ID15J3.pcapng Packets: 158 · Dropped: 0 (0.0%) Profile: Default

Masscan

On a side note, Masscan uses a similar approach to discover the available systems. However, to complete its network scan quickly, Masscan is quite aggressive in the rate at which it generates packets. The syntax is quite similar: `-p` can be followed by a port number, list, or range. Consider the following examples:

- `masscan 10.200.6.0/24 -p443`
- `masscan 10.200.6.0/24 -p80,443`
- `masscan 10.200.6.0/24 -p22-25`

Masscan is not installed on the `AttackBox`; however, it can be installed using `apt install masscan`.

Answer the questions below

Which TCP ping scan does not require a privileged account?

TCP SYN Ping

✓ Correct Answer

Which TCP ping scan requires a privileged account?

TCP ACK Ping

✓ Correct Answer

What option do you need to add to Nmap to run a TCP SYN ping scan on the telnet port?

-PS23

✓ Correct Answer



Nmap Live Host Discovery completed!

[sufriyahussain.r](#) completed another room on their cyber security journey.

Completed tasks

9

Points earned

160

Streak

2

Share on social media

Share your win before starting the next room

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Reddit